Scientific Computing 2026

Statistical estimation

Method of moments, maximum likelihood, and MAP

Lecture 5

Five observed waiting times

Independent waits between requests, in minutes:

$$1,\ 2,\ 3,\ 4,\ 5. \qquad \bar{t} = 3, \qquad \sum_i t_i = 15.$$

Keep the exponential model, but let its rate be unknown:

$$T_i \sim \mathrm{Exp}(\lambda), \qquad \lambda > 0.$$

Which value of λ should we report?

Chapter 4 studied variation under a specified model. Here the observations help us choose its parameter.

A statistical model

Definition: Statistical model

A family of possible observation laws

$$\{P_\theta : \theta \in \Theta\}.$$

The parameter space Θ specifies the allowed values.

Exponential rate: $\lambda \in (0, \infty)$.
Gaussian mean and variance: $(\mu, v) \in \mathbb{R} \times (0, \infty)$.

The family is an assumption about the observations. Estimating its parameters does not verify that assumption.

Estimator, estimate, and identifiability

Definition: Estimator and estimate

An estimator is a data-based rule $\widehat{\theta} = T(X_1, \ldots, X_n)$. Its realized value $T(x_1, \ldots, x_n)$ is an estimate.

For the waits: the rule $1/\bar{T}_n$ gives $1/3$ per minute.

Identifiability: different parameter values must give different laws. For $X \sim \mathcal{N}(a^2, 1)$, the data cannot distinguish a from $-a$.

Method of moments

Definition: Method of moments

Match selected model moments to their empirical counterparts:

$$\mathbb{E}_{\widehat{\theta}}[X^j] = \frac{1}{n}\sum_{i=1}^{n} x_i^j.$$

Solve for the unknown parameters within their allowed space.

For exponential waits,

$$\mathbb{E}_{\lambda}[T] = \frac{1}{\lambda} \quad \Longrightarrow \quad \widehat{\lambda}_{\mathrm{MM}} = \frac{1}{\bar{t}} = \frac{1}{3}.$$

Two moments for a Gamma model

Shape $\alpha > 0$, rate $\beta > 0$:

$$\mathbb{E}[X] = \frac{\alpha}{\beta}, \qquad \mathrm{Var}(X) = \frac{\alpha}{\beta^2}.$$

With $v_n = n^{-1} \sum_i (x_i - \bar{x})^2$, match

$$\bar{x} = \frac{\widehat{\alpha}}{\widehat{\beta}}, \qquad v_n = \frac{\widehat{\alpha}}{\widehat{\beta}^2}.$$

Result: Gamma moment estimates

$$\widehat{\alpha} = \frac{\bar{x}^2}{v_n}, \qquad \widehat{\beta} = \frac{\bar{x}}{v_n}, \qquad v_n > 0.$$

Matching the mean and spread

For $1, 2, 3, 4, 5$ minutes,

$$\bar{x} = 3, \qquad v_n = 2, \qquad \widehat{\alpha} = 4.5, \qquad \widehat{\beta} = 1.5 \text{ per minute.}$$

The Gamma fit matches both summaries.
The exponential fit has variance $\bar{x}^2 = 9$.

The denominator n follows from matching raw moments:

$$v_n = \widehat{m}_2 - \widehat{m}_1^2.$$

If $v_n = 0$, these Gamma moment equations have no finite positive solution.

Conditions for moment estimation

The selected population moments must exist and distinguish the parameters.

The LLN makes empirical moments approach population moments. A well-defined, continuous inverse then makes the parameter estimates converge.

Finite samples can still give no admissible solution, multiple solutions, or estimates that react sharply to small changes in the moments.

Likelihood

Definition: Likelihood for an i.i.d. sample

With the observations held fixed,

$$L(\theta; x) = \prod_{i=1}^{n} f_\theta(x_i).$$

Vary θ to compare candidate models for the recorded sample.

Discrete data use probability masses. Continuous data use densities.

Likelihood is not a distribution over θ. Dependent observations require their actual joint law.

Maximum likelihood and logarithms

Definition: Maximum likelihood estimator

$$\widehat{\theta}_{\mathrm{ML}} \in \arg\max_{\theta \in \Theta} L(\theta; x).$$

The maximizer may be nonunique or may fail to exist.

The increasing logarithm preserves the maximum:

$$\ell(\theta; x) = \sum_i \log f_\theta(x_i).$$

Drop only terms independent of the parameter. Adding logs also avoids multiplying many very small numbers.

The exponential rate MLE

For total observed waiting time $s > 0$,

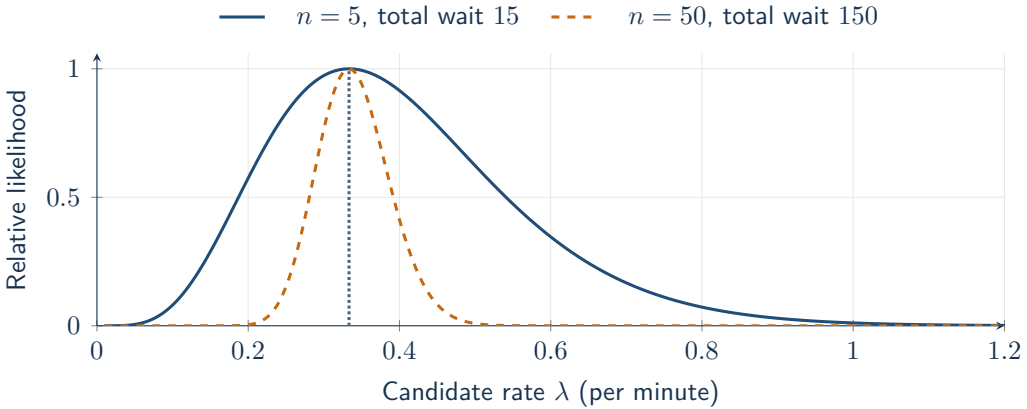
$$L(\lambda) = \lambda^n e^{-\lambda s}, \qquad \ell(\lambda) = n \log \lambda - \lambda s.$$

$$\ell'(\lambda) = \frac{n}{\lambda} - s, \qquad \ell''(\lambda) = -\frac{n}{\lambda^2} < 0.$$

Result: Unique maximum

$$\widehat{\lambda}_{\mathrm{ML}} = \frac{n}{s} = \frac{1}{\bar{t}}.$$

Likelihood distinguishes candidate rates



Two illustrative datasets have average wait three minutes. Both estimates are $1/3$ per minute. The vertical scale is $L(\lambda)/L(\widehat{\lambda})$.

A parameter-dependent support

For $X_i \sim \mathrm{Uniform}(0, \theta)$, let $x_{(n)} = \max_i x_i > 0$.

$$L(\theta) = \theta^{-n} \mathbf{1}_{\{\theta \geq x_{(n)}\}}.$$

An endpoint below an observation has likelihood zero.
Above the sample maximum, likelihood decreases.

Result: Uniform endpoint MLE

$$\widehat{\theta}_{\mathrm{ML}} = x_{(n)}.$$

The maximum lies at a boundary. No derivative-zero equation finds it.

Moment estimates and MLEs can differ

Uniform model, with observations $0.2, 0.4, 0.5, 0.7, 0.9$:

$$\widehat{\theta}_{\mathrm{MM}} = 2\bar{x} = 1.08, \qquad \widehat{\theta}_{\mathrm{ML}} = \max_i x_i = 0.9.$$

Moment matching uses $\mathbb{E}[X] = \theta/2$.
Likelihood also uses the support restriction.

A Gaussian mean estimate

For i.i.d. $X_i \sim \mathcal{N}(\mu, v)$ with $v > 0$,

$$\ell(\mu, v) = -\frac{n}{2}\log(2\pi) - \frac{n}{2}\log v - \frac{1}{2v}\sum_i (x_i - \mu)^2.$$

For fixed v, the identity

$$\sum_i (x_i - \mu)^2 = \sum_i (x_i - \bar{x})^2 + n(\mu - \bar{x})^2$$

puts the maximum at $\widehat{\mu} = \bar{x}$.

Here v denotes variance, not standard deviation.

A Gaussian variance estimate

Write $Q = \sum_i (x_i - \bar{x})^2 > 0$.

$$\frac{\partial \ell(\bar{x}, v)}{\partial v} = \frac{Q - nv}{2v^2}.$$

The derivative changes from positive to negative at $v = Q/n$.

Result: Gaussian MLEs

$$\widehat{\mu} = \bar{x}, \qquad \widehat{v} = \frac{1}{n} \sum_i (x_i - \bar{x})^2.$$

For identical observations, $Q = 0$ and the likelihood grows without bound as v approaches zero.

Bias and mean squared error

Repeat the entire sampling experiment with true parameter θ fixed.

Definition: Bias and MSE

$$\mathrm{Bias}_\theta(\widehat{\theta}) = \mathbb{E}_\theta[\widehat{\theta}] - \theta,$$
$$\mathrm{MSE}_\theta(\widehat{\theta}) = \mathbb{E}_\theta[(\widehat{\theta} - \theta)^2].$$

Unbiasedness means zero bias for every allowed parameter value.

Bias describes repeated-sample averages. One realized estimate has an error.

Bias and variance contribute to error

Write

$$\widehat{\theta} - \theta = \left(\widehat{\theta} - \mathbb{E}_\theta \widehat{\theta}\right) + \left(\mathbb{E}_\theta \widehat{\theta} - \theta\right).$$

Square and take expectations. The cross term has expectation zero.

Result: Bias–variance decomposition

$$\mathrm{MSE}_\theta(\widehat{\theta}) = \mathrm{Var}_\theta(\widehat{\theta}) + \mathrm{Bias}_\theta(\widehat{\theta})^2.$$

Under squared error, smaller variance can compensate for some bias.

The $n-1$ variance correction

For i.i.d. observations with finite variance,

$$\sum_i (X_i - \bar{X})^2 = \sum_i (X_i - \mu)^2 - n(\bar{X} - \mu)^2.$$

Taking expectations gives $n\sigma^2 - n(\sigma^2/n) = (n-1)\sigma^2$.

Result: Unbiased sample variance, $n \geq 2$

$$\mathbb{E}[\widehat{v}_{\mathrm{ML}}] = \frac{n-1}{n}\sigma^2, \qquad \mathbb{E}\left[\frac{1}{n-1}\sum_i (X_i - \bar{X})^2\right] = \sigma^2.$$

The expectation identity needs no Gaussian assumption. The Gaussian likelihood still chooses denominator n.

Squared error for a uniform endpoint

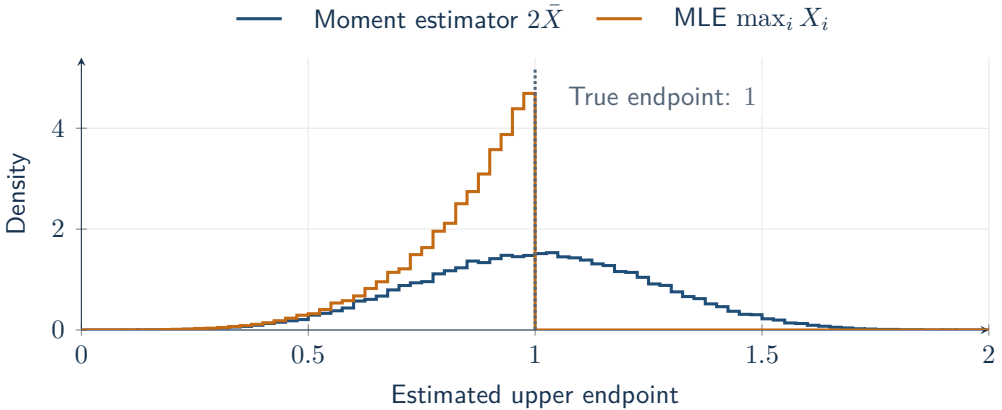
For five i.i.d. Uniform$(0, 1)$ observations:

Estimator	Bias	Mean squared error
$2\bar{X}$	0	$1/15 \approx 0.0667$
$\max_i X_i$	$-1/6$	$1/21 \approx 0.0476$

The maximum's smaller spread outweighs its bias in this comparison.

For $M = \max_i X_i$, use $\mathbb{P}(M \le m) = m^5$ on $[0, 1]$. Integrating its density gives the moments used in the table.

The distribution of the estimates



50,000 independent datasets of size five from Uniform$(0, 1)$. The MLE always underestimates the endpoint, but has smaller MSE here.

Consistency

Definition: Consistent estimator sequence

For every $\varepsilon > 0$,

$$\mathbb{P}_\theta(|\widehat{\theta}_n - \theta| > \varepsilon) \longrightarrow 0.$$

For exponential waits, the LLN gives $\bar{T}_n \to 1/\lambda$. Continuity of the reciprocal at this positive limit gives $1/\bar{T}_n \to \lambda$ in probability.

This estimator is consistent but biased: $\mathbb{E}[\widehat{\lambda}] = n\lambda/(n-1)$ for $n > 1$. Using only X_1 can be unbiased for a mean but inconsistent.

Gaussian regression gives squared error

Fix the features and assume conditionally independent responses:

$$Y_i \mid x_i \sim \mathcal{N}(f_\theta(x_i), \sigma^2), \qquad \sigma^2 > 0 \text{ known and shared.}$$

Result: Negative conditional log-likelihood

$$-\ell(\theta) = \frac{n}{2}\log(2\pi\sigma^2) + \frac{1}{2\sigma^2}\sum_i (y_i - f_\theta(x_i))^2.$$

The parameter-dependent term is a positive multiple of squared error. An optimization algorithm searches for parameters that minimize this sum.

Fitting a shared noise variance

Define the residual sum of squares:

$$\mathrm{RSS}(\theta) = \sum_i (y_i - f_\theta(x_i))^2.$$

For an unknown shared variance $v > 0$,

$$-\ell(\theta, v) = \frac{n}{2} \log v + \frac{\mathrm{RSS}(\theta)}{2v} + \text{constant}.$$

For fixed θ with $\mathrm{RSS}(\theta) > 0$,

$$\widehat{v}(\theta) = \frac{\mathrm{RSS}(\theta)}{n}.$$

A perfect fit has RSS $= 0$ and allows variance collapse. A finite fit then needs an additional constraint or a different model.

Predicting a different variance for each input

Let $\mu_i = \mu_\theta(x_i)$ and $v_i = v_\theta(x_i) > 0$.

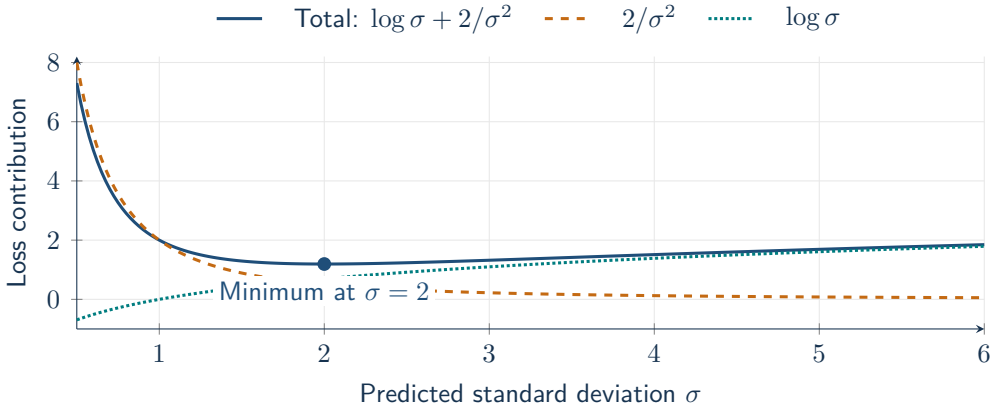
Result: Gaussian loss with input-dependent variance

$$-\ell(\theta) = \frac{1}{2} \sum_i \left[\log v_i + \frac{(y_i - \mu_i)^2}{v_i} \right] + \text{constant}.$$

A larger variance reduces the weighted residual but increases $\log v_i$.

$v_i = e^{s_i}$ enforces positivity. It does not prevent collapse at a perfectly fitted observation.

The variance term has a cost



A fixed residual $y - \mu = 2$ gives loss $\log\sigma + 2/\sigma^2$. Omitting $\log\sigma$ would reward arbitrarily large predicted noise.

Bernoulli responses give binary log loss

For $Y_i \in \{0, 1\}$, predict $p_i = p_\theta(x_i) \in (0, 1)$.

Result: Negative Bernoulli log-likelihood

$$-\ell(\theta) = -\sum_i \Big[y_i \log p_i + (1 - y_i) \log(1 - p_i) \Big].$$

For a positive response:

$$-\log 0.9 \approx 0.105, \qquad -\log 0.1 \approx 2.303.$$

Confident wrong predictions incur a large loss.

A prior distribution over the parameter

Specify prior density $\pi(\theta)$ before using the current observations. The model supplies $L(\theta; x)$ conditional on each candidate parameter.

$$\pi(\theta \mid x) = \frac{L(\theta; x)\pi(\theta)}{\int_{\Theta} L(u; x)\pi(u)\, du}.$$

The denominator normalizes the posterior and is constant in θ.

The denominator must be positive and finite. For a discrete parameter, replace the integral by a sum.

Maximum a posteriori estimation

Definition: MAP estimate

Maximize the posterior density, or posterior mass for a discrete parameter:

$$\widehat{\theta}_{\mathrm{MAP}} \in \arg\max_{\theta \in \Theta} \{\ell(\theta; x) + \log \pi(\theta)\}.$$

Equivalently, minimize negative log-likelihood plus negative log-prior.

MAP summarizes the posterior by its mode. It does not report the whole posterior or its uncertainty.

A prior for a Bernoulli probability

The Beta family places a density on $0 < p < 1$:

$$\pi(p) = \frac{p^{a-1}(1-p)^{b-1}}{B(a,b)}, \qquad a, b > 0,$$

$$B(a,b) = \int_0^1 u^{a-1}(1-u)^{b-1}\, du.$$

The chosen a, b describe the prior.
Beta$(2,2)$ has density $6p(1-p)$ and favors values near $1/2$.

Updating a Beta prior

For k successes in n independent Bernoulli trials,

$$L(p)\pi(p) \;\propto\; p^{k+a-1}(1-p)^{n-k+b-1}.$$

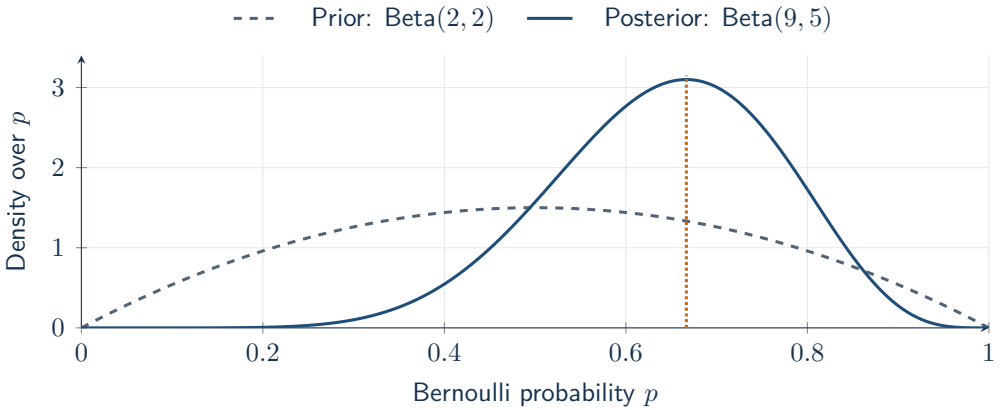
Thus $p \mid x \sim \mathrm{Beta}(a+k, b+n-k)$.

Result: Interior posterior mode

$$\widehat{p}_{\mathrm{MAP}} = \frac{k+a-1}{n+a+b-2}.$$

This formula requires $a+k>1$ and $b+n-k>1$.

Seven successes in ten trials



Beta$(2,2)$ prior, Beta$(9,5)$ posterior. MLE $= 0.7$ and MAP $= 2/3$; the dotted line marks the posterior mode.

Posterior mode and posterior mean

For the Beta$(9, 5)$ posterior,

$$\text{MAP} = \frac{2}{3} \approx 0.667, \qquad \mathbb{E}[p \mid x] = \frac{9}{14} \approx 0.643.$$

The mode maximizes density. The mean minimizes posterior expected squared error, by the conditional-mean result from Chapter 3.

For continuous parameters, MAP depends on the parameterization: transforming a density introduces a Jacobian factor.

A Gaussian prior for an unknown mean

Observation model: $X_i \sim \mathcal{N}(\mu, \sigma^2)$, known $\sigma^2 > 0$. Prior: $\mu \sim \mathcal{N}(m_0, \tau^2)$, with $\tau^2 > 0$.

$$-\log \pi(\mu \mid x) = \frac{1}{2\sigma^2} \sum_i (x_i - \mu)^2 + \frac{(\mu - m_0)^2}{2\tau^2} + \text{constant}.$$

Result: Precision-weighted estimate

$$\widehat{\mu}_{\mathrm{MAP}} = \frac{(n/\sigma^2)\bar{x} + (1/\tau^2) m_0}{n/\sigma^2 + 1/\tau^2}.$$

How much weight the prior receives

Precision means inverse variance. The sample mean has variance σ^2/n, so its precision is n/σ^2.

$$n = 4, \quad \sigma^2 = 4, \quad \bar{x} = 1.6, \quad m_0 = 0, \quad \tau^2 = 1.$$

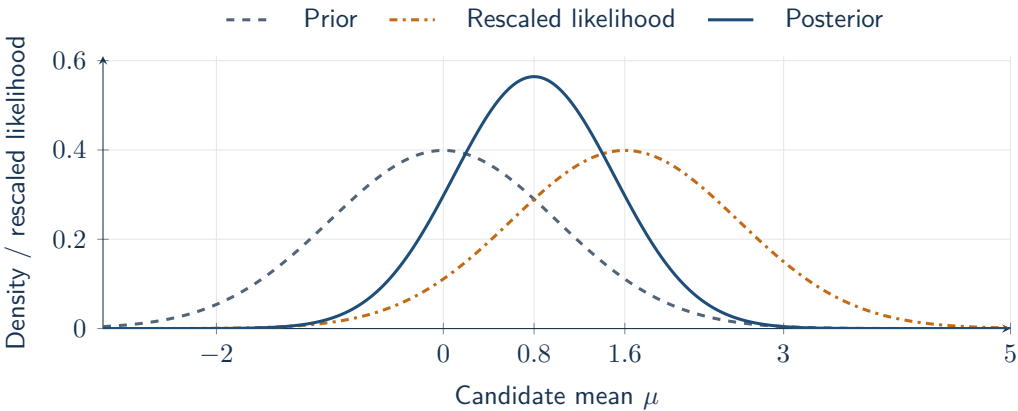
Both precisions are one:

$$\widehat{\mu}_{\mathrm{ML}} = 1.6, \qquad \widehat{\mu}_{\mathrm{MAP}} = 0.8.$$

The posterior is $\mathcal{N}(0.8, 0.5)$.

Here the posterior mean and mode coincide. With the prior fixed, more observations increase the relative data weight.

The Gaussian posterior



The likelihood curve has been rescaled to integrate to one for comparison. That rescaling does not make it a prior or posterior.

A Gaussian prior gives a quadratic penalty

Gaussian responses with known variance σ^2 and prior $w \sim \mathcal{N}(0, \tau^2 I)$ give

$$-\log \pi(w \mid x, y) = \frac{\mathrm{RSS}(w)}{2\sigma^2} + \frac{\|w\|^2}{2\tau^2} + \text{constant}.$$

Result: Equivalent MAP objective

$$\mathrm{RSS}(w) + \alpha \|w\|^2, \qquad \alpha = \frac{\sigma^2}{\tau^2}.$$

For a linear model, this is ridge regression. The linear-algebra solution comes later in the course.

Loss normalization changes the penalty coefficient

The same prior gives equivalent objectives

$$\sum_i (y_i - f_w(x_i))^2 + \alpha \|w\|^2$$

and

$$\frac{1}{n} \sum_i (y_i - f_w(x_i))^2 + \frac{\alpha}{n} \|w\|^2.$$

Replacing a sum by an average without adjusting the coefficient changes the relative penalty strength.

An unpenalized intercept requires a different prior for that coefficient.

Three point-estimation rules

Method	Construction
Method of moments	Match selected model and empirical moments
Maximum likelihood	Maximize the observation likelihood
MAP	Maximize likelihood times a specified prior

Their estimates can agree, but their definitions differ.

Point estimates alone do not quantify uncertainty. Chapter 6 develops sampling distributions, confidence intervals, and bootstrap.